

(a)

$$\begin{aligned} f(x) &= \frac{1}{2}x'Ax + b'x \\ &= \frac{1}{2}\sum_{i=1}^n\sum_{j=1}^na_{ij}x_ix_j + \sum_{i=1}^nb_ix_i \end{aligned}$$

求偏导数可得

$$\begin{aligned} \frac{\partial f}{\partial x_i} &= \frac{1}{2}\left(2a_{ii}x_i + \sum_{j \neq i} 2a_{ij}x_j\right) + b_i \\ &= \sum_{j=1}^na_{ij}x_j + b_i \\ &= (a_{i1}, a_{i2}, \dots, a_{in})x + b_i \end{aligned}$$

于是

$$\nabla f(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix} = Ax + b$$

(b)

$$\nabla f(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix} = \begin{bmatrix} g'[h(x)] \frac{\partial h}{\partial x_1} \\ g'[h(x)] \frac{\partial h}{\partial x_2} \\ \vdots \\ g'[h(x)] \frac{\partial h}{\partial x_n} \end{bmatrix} = g'[h(x)] \cdot \nabla h(x)$$

(c)

已知

$$\frac{\partial f}{\partial x_i} = \sum_{k=1}^na_{jk}x_k + b_j$$

再次求导可得

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial}{\partial x_i} \left(\frac{\partial f}{\partial x_j} \right) = a_{ji}$$

此时 $H_{ij} = a_{ij}$, 所以 $H = A' = A$